

2.4 Boolean logic

Where this unit sits, what it covers, and how far

Where this unit sits

Qualification	GCSE Computer Science (9-1), specification code J277, from 2020
Component	Component 02: Computational thinking, algorithms and programming — 80 marks, 1 hour 30 minutes
What OCR says the component covers	computational thinking: algorithms, programming techniques, producing robust programs, computational logic and translators
This unit	topic 2.4 Boolean logic
Checked	3 October 2026, at https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/

How far this has been checked. The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

Before it: 1.1 Systems architecture, 1.2 Memory and storage, 1.3 Computer networks, connections and protocols, 1.4 Network security, 1.5 Systems software, 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs

After it: 2.5 Programming languages and Integrated Development Environments, PY Python programming

The lesson sequence and the questions it answers

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
1	Logic gate lab b1-101	How do you draw a logic diagram?	15	24
2	Truth table detectives b1-102	How do you complete a truth table?	11	38
3	Circuit workshop b1-103	How do you create logic diagrams from truth tables?	14	35
4	End-of-topic test b1-104-test	assesses the whole unit	0	0
5	Logic sprint b1-bonus	stretch work, no new content	1	0

Prior knowledge, vocabulary and skills

Assumed before the unit: the outcomes of 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 2.2, 2.3

Vocabulary introduced: AND gate, Logic gate, Truth table, Boolean operator, Logic diagram, Intermediate output

What a pupil does, not only reads: circuit, expr, match, mcq, sprint, table

Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

How this was worked out. The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage.json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know the symbols for the logic gates AND, OR and NOT (b1-l01-o1)	bl-l01	yes	yes	yes	Full
understand the truth tables for AND, OR and NOT (b1-l01-o2)	bl-l01	yes	yes	yes	Full
combine Boolean operators using AND, OR and NOT (b1-l01-o3)	bl-l01	yes	yes	yes	Full
understand how to complete truth tables from one- and two-level logic diagrams (b1-l02-o1)	bl-l02	yes	yes	yes	Full
understand how to create, complete or edit logic diagrams and truth tables for given scenarios (b1-l03-o1)	bl-l03	yes	yes	yes	Full

5 full · 0 partial · 0 not covered, of 5 outcomes.

Gaps, assumptions and overlap

None found by the trace.

Assessed, but matching no outcome this unit states

Each of these is a real exam question on a worksheet in this unit. Either the outcome it tests is worded differently, or it is genuinely assessing something the unit does not claim to teach.

- b1-l01-x2 Give the output of an OR gate when its inputs are 0 and 1.

What could not be traced

The wall text of b1 - bonus is painted into the 360 image and held nowhere as text, so an outcome in that lesson lesson can only be traced through its activities and its exam questions. Read the rows above for it with that in mind.

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